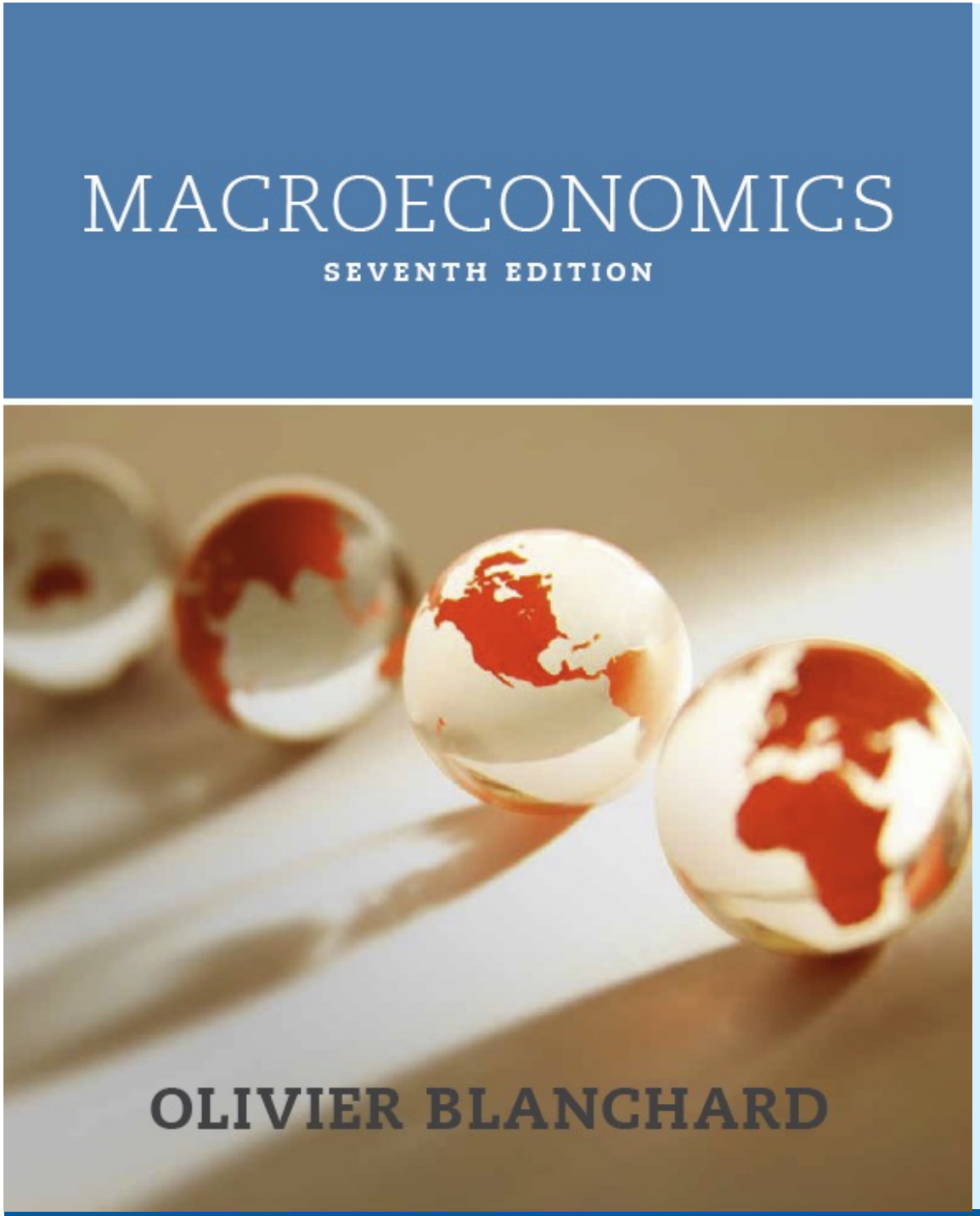


The Goods Market in an Open Economy

Chapter 18

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Chapter 18 Outline

The Goods Market in an Open Economy

1. The *IS* Relation in the Open Economy
2. Equilibrium Output and the Trade Balance
3. Increases in Demand—Domestic or Foreign
4. Depreciation, the Trade Balance, and Output
5. Saving, Investment, and the Current Account Balance

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1. The *IS* Relation in the Open Economy

- In an open economy, the **demand for domestic goods** is

$$Z = C + I + G - IM/\varepsilon + X \qquad (18.1)$$

where *C*, *I*, and *G* constitute the total **domestic demand for goods**, domestic or foreign.

where $\varepsilon = \dfrac{EP}{P^*}$

Note: The **Real Exchange Rate** is the ratio of the dollar price of a basket of goods and services in the home country divided by the dollar price of the same basket of goods and services in a foreign country.

$$\text{Domestic demand:} \quad C + I + G = \underset{(+)}{C(Y - T)} + \underset{(+,-)}{I(Y, r)} + G$$

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Nominal versus Real exchange rates

- The nominal rate converts currencies.
- The real rate compares the prices of goods across countries.

NOMINAL EXCHANGE RATE

E = 0.90 €/US$

One dollar costs €0.90

REAL EXCHANGE RATE

$q = E \times P^* / P$

Price of foreign goods relative to domestic goods

EXAMPLE: EURO AREA IS THE HOME ECONOMY

US basket	Convert at E	US basket in euros	Euro-area basket
$120	**× 0.90**	**€108**	**€100**

q = €108 / €100 = 1.08

US goods are 8% more expensive than euro-area goods. Euro-area goods are relatively competitive.

Quotation: euros per dollar. With this convention, a rise in E or q means a depreciation of the euro.

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1. The *IS* Relation in the Open Economy

- Imports is positively related to domestic income (Y) and the real exchange rate (ε):

$$IM = IM(Y, \varepsilon) \qquad (18.2)$$
$$(+, +)$$

- Exports is positively related to foreign income (Y^*) and negatively related ε:

$$X = X(Y^*, \varepsilon) \qquad (18.3)$$
$$(+, -)$$

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2. Equilibrium Output and the Trade Balance

- Because $Y = Z$, the equilibrium condition for output can be expressed as:

$$Y = C(Y - T) + I(Y, r) + G - IM(Y, \varepsilon)/\varepsilon + X(Y^*, \varepsilon)$$

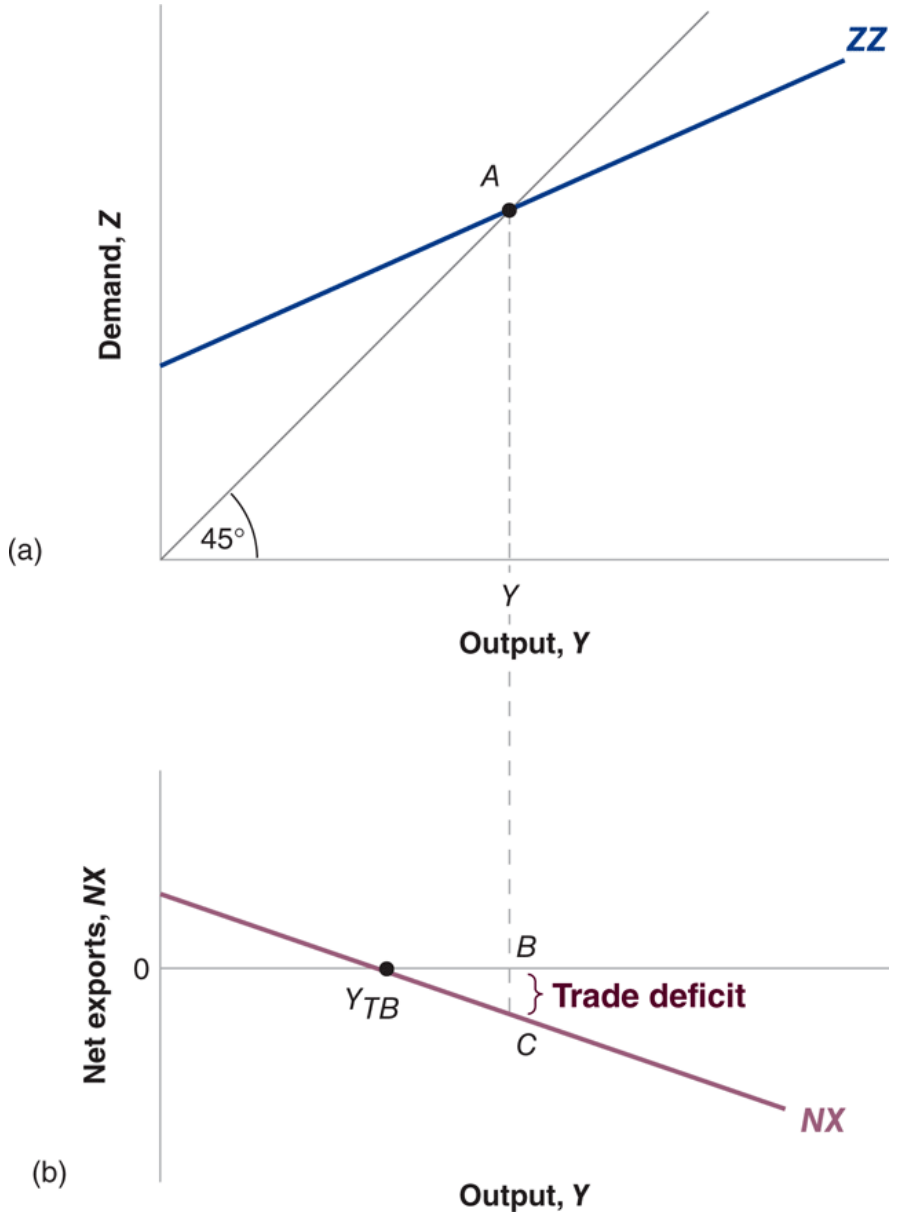
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2. Equilibrium Output and the Trade Balance

Figure 18-2 Equilibrium Output and Net Exports

The goods market is in equilibrium when domestic output is equal to the demand for domestic goods.

At the equilibrium level of output, the trade balance may show a deficit or a surplus.



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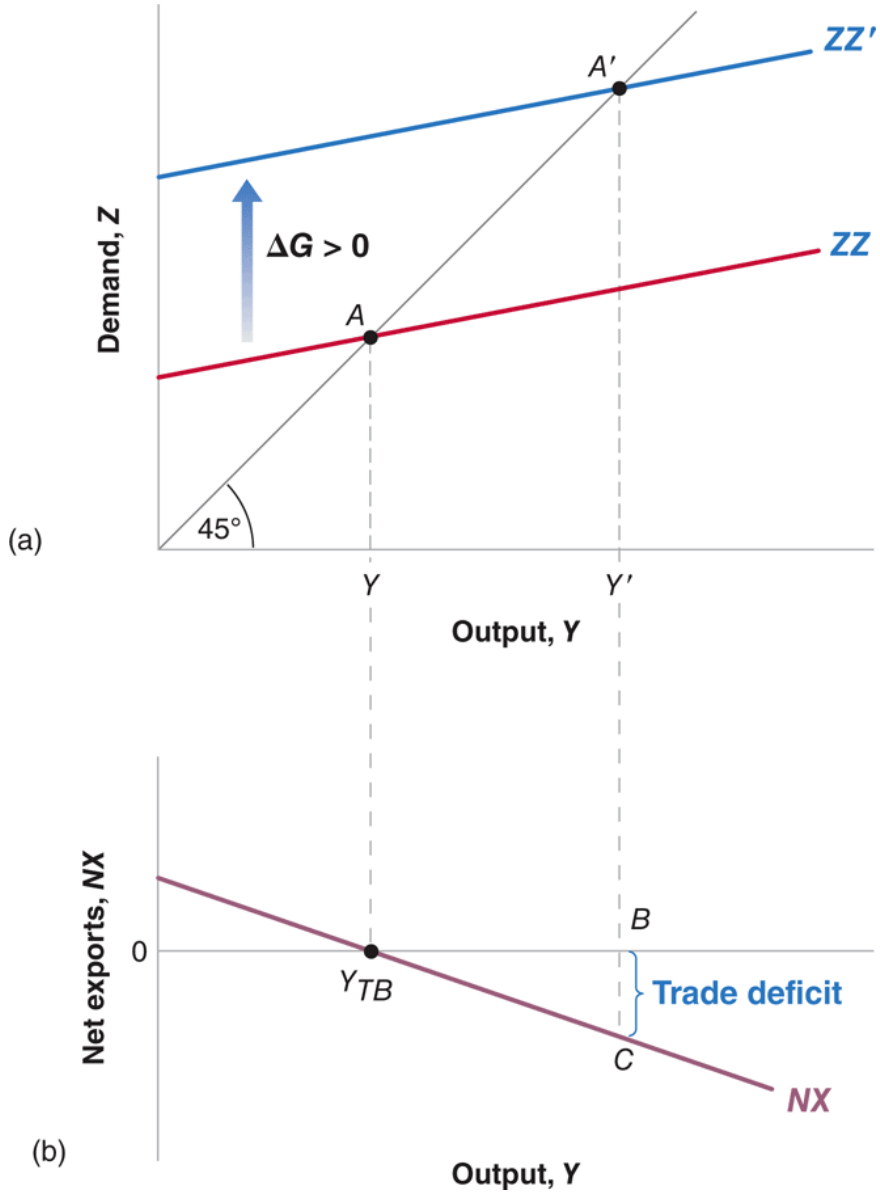
3. Increases in Demand– Domestic or Foreign

- Differences with a closed economy:
  - Effect on trade balance: An increase in output leads to a *trade deficit*
  - Smaller effect of government spending on output: Because *ZZ* is flatter than *DD*, the ***multiplier is smaller in the open economy***

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3. Increases in Demand–Domestic or Foreign

Figure 18-3 The Effects of an Increase in Government Spending



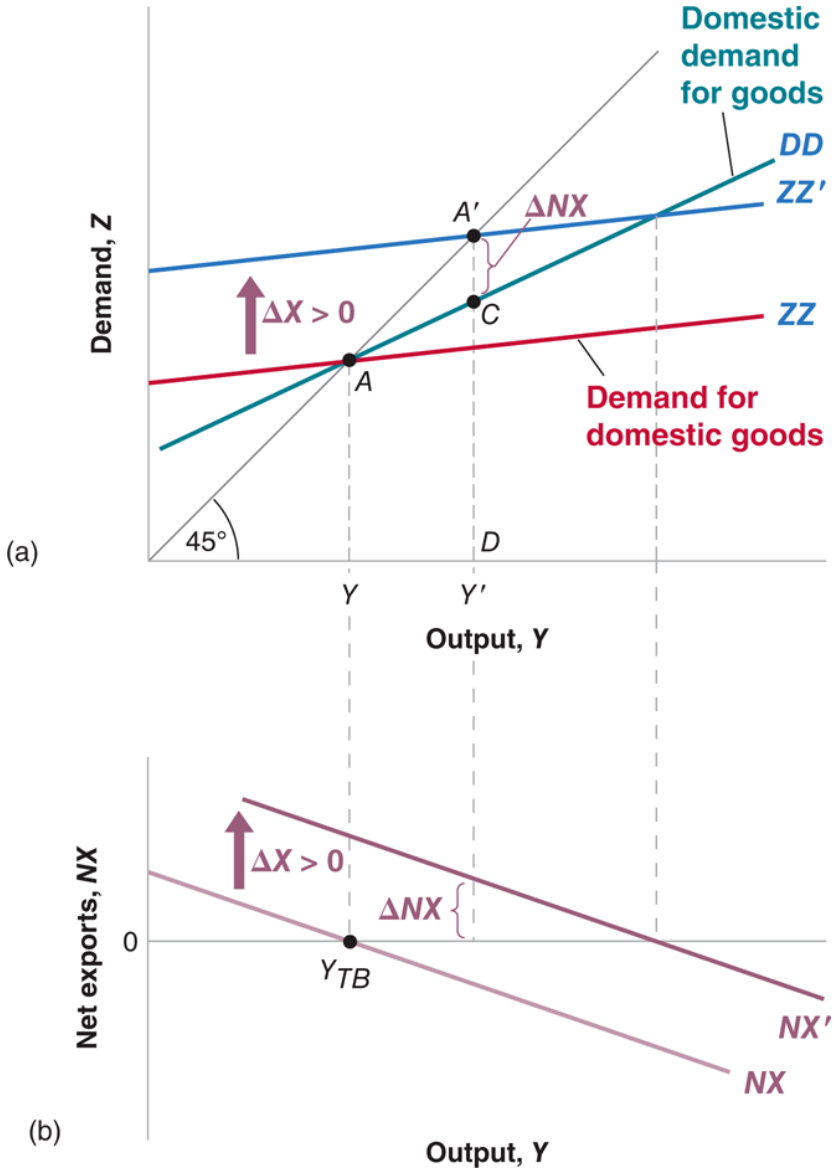
An increase in government spending leads to

- an increase in output
- a trade deficit.

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3. Increases in Demand–Domestic or Foreign

Figure 18-4 The Effects of an Increase in Foreign Demand



An increase in foreign demand leads to

- an increase in output
- a trade surplus.

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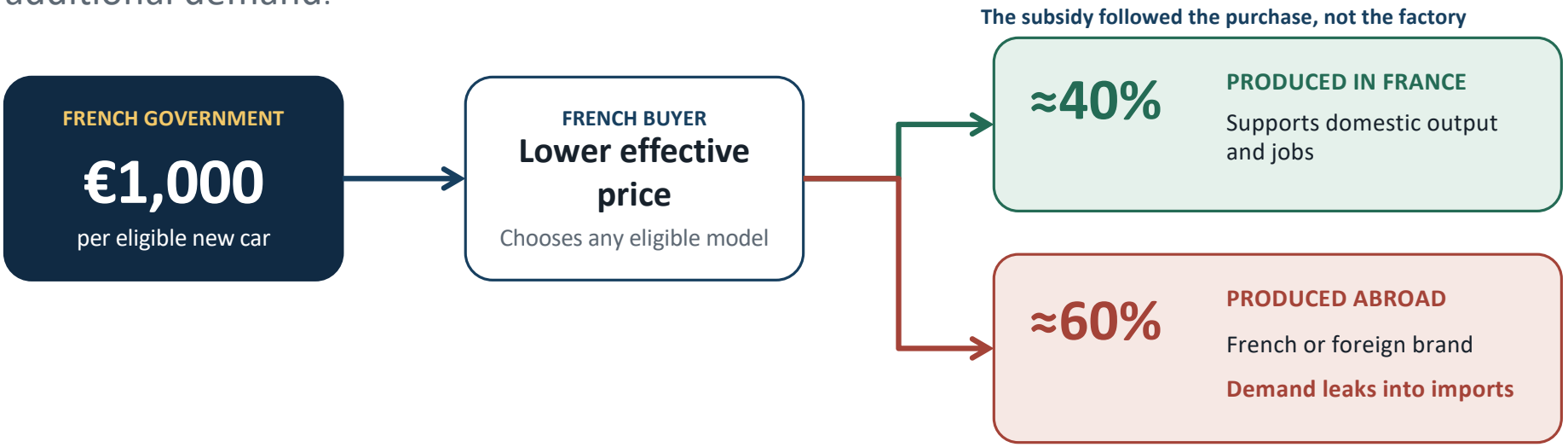
3. Increases in Demand– Domestic or Foreign

- Differences with a closed economy:
  - An **increase in domestic demand** leads to an increase in domestic output but leads also to a **deterioration of the trade balance**.
  - An **increase in foreign demand** leads to an increase in domestic output and an **improvement in the trade balance.**
  - Effect on trade balance: **An increase in output leads to a *trade deficit***
  - Smaller effect of government spending on output: **the *multiplier is smaller in the open economy***

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France’s car subsidy and import leakage

The 2009 subsidy boosted car purchases, but production abroad captured much of the additional demand.



$$Y = C + I + G + X - M$$

An imported car raises consumption (C) and imports (M) together.

Only the value created in France adds directly to French GDP.

FACT CHECK German brands did not dominate subsidised registrations: Volkswagen 4.2%, Opel 4.3%.

Source: French Ministry of Ecological Transition, May 2010

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3. Increases in Demand– Domestic or Foreign

- Implications:
  - Shocks to demand in one country affect all other countries.
  - Economic interactions complicate the task of policy makers. **Policy coordination** is not so easy to achieve.

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4. Depreciation, the Trade Balance, and Output

- Recall the real exchange rate equation:

$$\varepsilon = \frac{EP}{P^*}$$

and the definition of net exports: $NX = X - IM/\varepsilon$

- Given the expressions from equations (18.2) and (18.3):

$$NX = X(Y^*, \varepsilon) - IM(Y, \varepsilon)/\varepsilon$$

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4. Depreciation, the Trade Balance, and Output

- Real depreciation affects the trade balance through three separate channels:

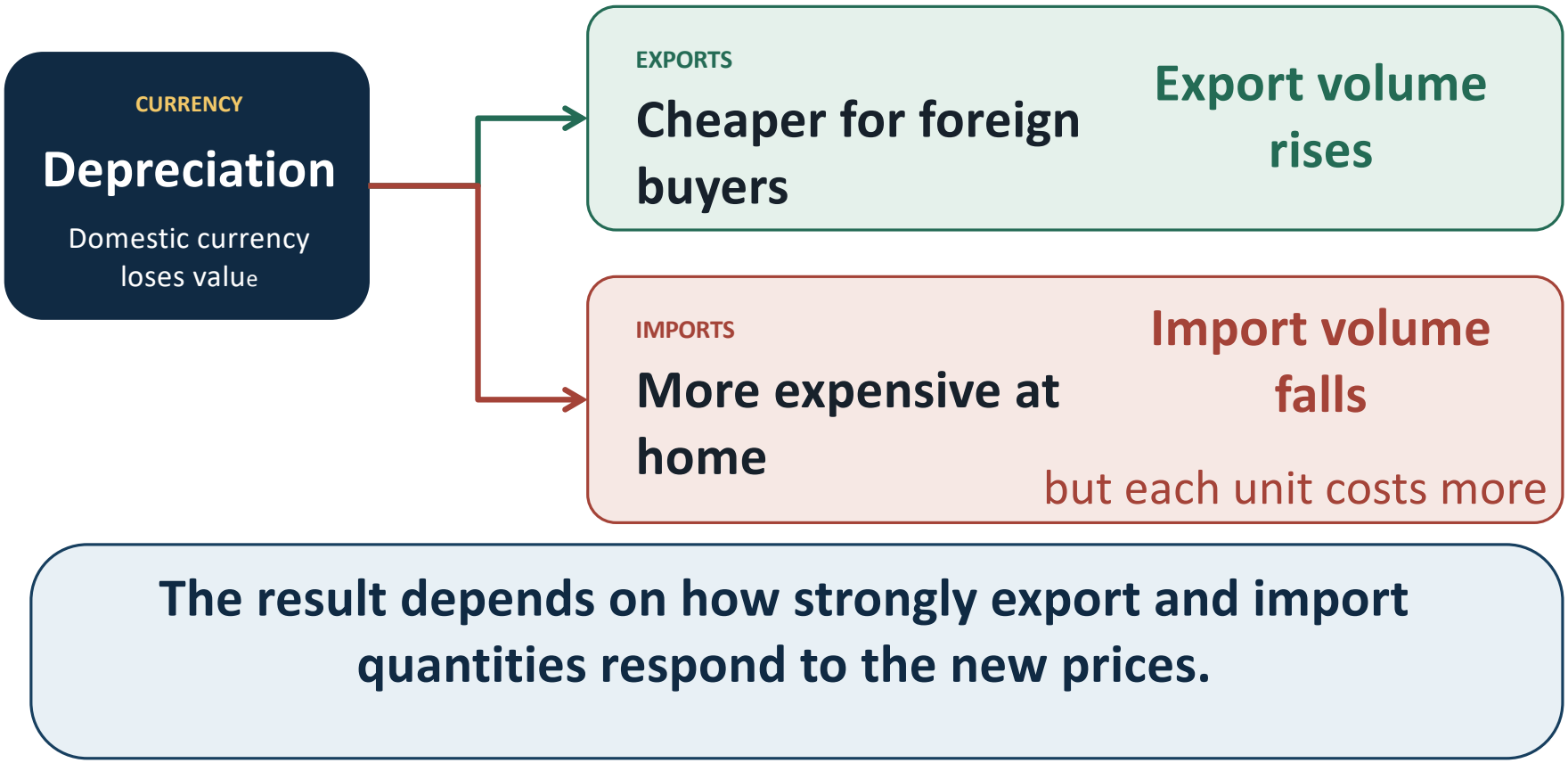
  1. *Exports, X, increase.*
  2. *Imports, IM, decrease.*
  3. *The relative price of foreign goods in terms of domestic goods, $1/\varepsilon$, increases.*

- **Marshall-Lerner condition**: A real depreciation leads to an increase in net exports.
- *Depreciation leads to a shift in demand, both foreign and domestic, toward domestic goods. The shift in demand leads, in turn, to both an increase in domestic output and an improvement in their trade balance.*

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The Marshall–Lerner condition

When does a currency depreciation improve the trade balance?



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5. Saving, Investment, and the Current Account Balance

- The equilibrium condition:

$$Y = C(Y - T) + I(Y, r) + G - IM(Y, \varepsilon)/\varepsilon + X(Y^*, \varepsilon)$$

- Move C to the left side, subtract taxes T from both sides and denote Net Exports:

$$Y - T - C = I + (G - T) + NX$$

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5. Saving, Investment, and the Current Account Balance

- Denote the current account by *CA*, so that the above equation becomes:

$$S = I + (G - T) + CA$$

meaning that the current account balance equal to saving minus investment:

$$CA = S + (T - G) - I$$

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5. Saving, Investment, and the Current Account Balance

- An increase in investment must be reflected either an increase in private saving or public saving, or in a deterioration of the current account balance.

- A deterioration in the government budget balance must be reflected in an increase in either private saving, or in a decrease in investment, or else in a deterioration of the current account balance (net exports).

- A country with a high saving rate must have either a high investment rate or a large current account surplus.

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Main Takeaways: Short Questions

1. What is the demand for domestic goods in an open economy?
2. What is the domestic demand of goods in an open economy?
3. What is the output condition of equilibrium in an open economy?
4. What is the effect of an increase of government expenditure on the trade balance?
5. What is the effect of an increase in foreign demand on the trade balance?
6. How does RER affect the trade balance?
7. What is the Marshal-Lerner condition?
8. What is the effect of an increase in the government deficit on savings, investment and the trade balance?

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